Internet of Things (IoT) with Applications

Undergraduate Engineering Course

Module 5

IoT Applications

Seven Domains Where Sensing, Networking, Computing, and Security Converge

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Unit V: IoT Applications

Concepts

Domains Covered:

Smart Agriculture · Healthcare IoT · Industrial IoT (IIoT)
Smart City · Smart Energy · Smart Transportation · Smart Home

Plus: a cross-domain synthesis of Modules 1–4

1 Introduction

Modules 1 through 4 developed the vocabulary, technologies, and principles of IoT one layer at a time: devices and reference architectures (Module 1), communication links and protocols (Module 2), computing platforms across the edge-to-cloud spectrum (Module 3), and security and analytics (Module 4). This final module does not introduce new theory; instead it shows how every one of those pieces is *selected and combined* in seven real-world application domains, and why the specific choices made in each domain follow directly from that domain's particular constraints.

A consistent pattern will recur throughout this module, matching the generic pipeline introduced in Module 3: **Sensors → Gateway/Network → Edge/Cloud Processing → Dashboard/Analytics → Actuator**. What changes from domain to domain is not this overall shape, but the specific technology chosen at each stage – which sensors, which communication technology from Module 2, how much processing happens at the edge versus the cloud, and which security and analytics priorities from Module 4 matter most. Reading this module with that lens – asking, for each domain, *why* a particular choice was made – is more valuable than memorising the application lists themselves.

2 Smart Agriculture

Smart agriculture applies IoT sensing and automation to monitor and manage crops, soil, and livestock across areas that are often large, remote, and short of both power and network infrastructure – constraints that shape nearly every technology choice in this domain.

- **Soil moisture monitoring:** Buried or surface soil-moisture sensors report water content, the primary input to automated irrigation decisions.
- **Automatic irrigation:** An actuated valve or pump is triggered, either by a simple threshold rule or by a cloud-based model, when soil moisture falls below a crop-specific target.
- **Crop and weather monitoring:** Cameras and weather stations track crop growth stage and local conditions (rainfall, temperature, humidity) that influence irrigation and treatment decisions.
- **Disease detection:** Computer vision (Module 3, Section 5.3) applied to leaf or crop imagery flags early signs of disease, often before visible symptoms would be obvious to a human inspector.
- **Smart fertilizer management:** Sensor-informed, variable-rate application of fertilizer matches nutrient delivery to actual soil and crop need rather than a uniform field-wide schedule.

Representative Pipeline

Soil/Weather Sensors → LoRaWAN Gateway (Module 2: long range, unlicensed spectrum, very low power suits remote fields) → Cloud Analytics (Module 3: irrigation-scheduling model) → Farmer Dashboard → Automated Irrigation Valve.

Because fields are large and often outside cellular coverage, **LoRaWAN** (Module 2, Section 3.1) is typically the preferred long-range technology here, and because irrigation decisions are not usually millisecond-critical, most processing can comfortably live in the cloud rather than at the edge – a different balance from the industrial and transportation domains discussed later in this module.

3 Healthcare IoT

Healthcare IoT connects wearable and bedside medical sensors to remote monitoring systems, prioritising two concerns above all others: data **confidentiality** (Module 4, Section 2), given the sensitivity of health data, and **reliability**, since a missed or delayed alert can have direct patient-safety consequences.

- **Remote patient monitoring:** Continuous or periodic vital-sign readings are transmitted from a patient's home or ward bed to a care team without requiring an in-person visit for every check.
- **ECG monitoring:** Wearable electrocardiogram sensors detect abnormal heart rhythms, often using an on-device or edge model (Module 3, Section 4.4) to flag events in real time.
- **SpO_2 and temperature monitoring:** Pulse-oximetry and temperature sensors track blood-oxygen saturation and body temperature, two of the most common vital signs monitored remotely.
- **Smart medical devices:** Connected infusion pumps, inhalers, and glucose monitors report usage and readings automatically, reducing manual logging errors.
- **Elderly-care systems:** Motion and fall-detection sensors support ageing patients living independently, alerting caregivers to falls or unusual inactivity.
- **Emergency alerts:** Threshold- or model-based alerts (Module 3, Section 5.2) notify care staff immediately when a monitored vital sign crosses a dangerous threshold.

Representative Pipeline

Wearable ECG/SpO_2 Sensor → BLE (Module 2: very low power, short range suits body-worn devices) → Smartphone/Gateway → Encrypted Cloud Storage (Module 4: confidentiality, access control) → Care-Team Dashboard + Emergency Alert.

BLE (Module 2, Section 2.1) dominates the device-to-gateway link here because wearable sensors are small, battery-powered, and worn close to a smartphone or bedside hub. End-to-end encryption and strict access control (Module 4, Sections 2.3 and 3.3) are non-negotiable given the sensitivity of health data, and emergency alerting logic is frequently pushed to the edge (the wearable or paired phone) precisely so that a life-threatening event is detected in milliseconds rather than waiting on a cloud round trip – the same latency argument made generally in Module 3, Section 4.5.

Worked Example

Justifying a design choice. A hospital wants a wearable ECG monitor's abnormal- rhythm alert to reach a nurse within two seconds of onset, but the hospital's Wi-Fi network occasionally experiences brief outages. Should abnormal-rhythm detection run in the cloud or at the edge (on the wearable or a bedside hub), and why?

Solution. At the edge. A cloud-only design would fail to alert during a Wi-Fi outage and could also exceed the two-second latency requirement even when connected; running detection locally on the wearable or bedside hub (Module 3, Section 4.4, Edge AI) allows an alert to be generated immediately and independently of network availability, with the cloud connection used only to forward the alert onward once restored.

4 Industrial IoT (IIoT)

Industrial IoT applies sensing and analytics to manufacturing and heavy-industry equipment, where the dominant concerns are **uptime** and **safety** rather than the power-constrained, remote-deployment concerns of Section 2.

- **Machine monitoring:** Vibration, temperature, and current sensors on industrial equipment provide continuous visibility into machine health.
- **Predictive maintenance:** As introduced in Module 3, Section 5.3, models trained on historical sensor data ahead of past failures forecast when a machine is likely to need maintenance, scheduling it before an unplanned breakdown occurs.
- **Asset tracking:** RFID or BLE tags (Module 2, Sections 2.3 and 2.1) track tools, parts, and equipment as they move through a facility or supply chain.
- **Industrial automation:** Sensor data drives automated control of production lines, often through the same real-time, edge-based logic discussed in Module 3, Section 4.3.
- **Digital twins:** A **digital twin** is a continuously updated virtual model of a physical machine or process, fed by live sensor data, used to simulate, monitor, and predict behaviour without directly experimenting on the physical equipment itself.
- **Quality monitoring:** Computer-vision or sensor-based inspection (Module 3, Section 5.4) detects manufacturing defects automatically, often faster and more consistently than manual inspection.

Representative Pipeline

Vibration/Current Sensors → Wired/Industrial Wi-Fi → Edge AI at Gateway (Module 3, Section 4.4: local anomaly detection for immediate shutdown) → Cloud (predictive- maintenance model, digital twin) → Maintenance Dashboard.

This is the domain that most clearly illustrates Module 3's edge-versus-cloud trade-off (Section 4.2): a safety-critical shutdown decision must run at the edge, within milliseconds, exactly as argued in Module 3's motor-shutdown worked example, while the longer-horizon predictive-maintenance model that forecasts *when* a bearing will fail weeks from now can run comfortably in the cloud, where far more historical data and computational power are available.

5 Smart City

Smart city applications operate at large, fixed-infrastructure scale – thousands to hundreds of thousands of largely stationary devices spread across a municipality – making the network-scaling concerns of Module 2 (IPv6, Section 5.1) and the fog-computing tier of Module 3 (Section 4.3) particularly relevant.

- **Smart traffic management:** Sensors and cameras at intersections adjust signal timing dynamically based on real-time traffic flow rather than a fixed schedule.
- **Smart parking:** Occupancy sensors in parking spaces feed a live map of available spaces to a driver-facing app, reducing time spent searching for parking.
- **Smart streetlights:** Lights dim or brighten automatically based on ambient light and detected pedestrian/vehicle presence, reducing energy use without compromising safety.

- **Smart waste management:** Fill-level sensors in waste bins let collection routes be planned based on actual need rather than a fixed schedule, reducing unnecessary collection trips.
- **Air-quality monitoring:** A distributed network of air-quality sensors provides fine-grained pollution data across a city, informing both public alerts and longer-term policy.
- **Water management:** Sensors across the water distribution network detect leaks, monitor usage, and support demand forecasting.

Representative Pipeline

Distributed Fixed Sensors (parking, air quality, waste bins) → NB-IoT / Zigbee Mesh (Module 2: massive device count, static locations) → Fog Nodes at District Level (Module 3, Section 4.3: local aggregation) → City Cloud Platform → Public Dashboards / Operations Centre.

The sheer device count in a city-scale deployment is exactly the scaling problem Module 2's discussion of **IPv6** and **fog computing** anticipates: routing every reading from tens of thousands of sensors directly to a central cloud would be both bandwidth- wasteful and slow, so district-level fog nodes aggregate and pre-filter data first, echoing the smart-building worked example from Module 3.

6 Smart Energy

Smart energy applications monitor and manage electricity generation, distribution, and consumption, with a strong analytics component (Module 3, Section 5) driving demand forecasting and grid stability.

- **Smart meters:** Digital meters report household or facility electricity consumption at fine time resolution, replacing manual monthly meter readings.
- **Smart grid:** A broader network of sensors and controls across the electricity distribution network enables automated fault detection, load balancing, and integration of distributed generation sources.
- **Energy monitoring:** Sub-metering at the circuit or appliance level gives finer-grained visibility into where energy is actually consumed within a building.
- **Demand prediction:** Predictive analytics (Module 3, Section 5.3) forecasts short- and long-term electricity demand, informing generation scheduling and pricing.
- **Renewable-energy monitoring:** Sensors on solar panels and wind turbines track generation output and equipment health, applying the same predictive-maintenance techniques used in Section 4 of this module.
- **Building energy management:** Building-wide systems combine occupancy, temperature, and lighting sensors to automatically optimise HVAC and lighting energy use.

Representative Pipeline

Smart Meter → NB-IoT / LTE-M (Module 2: static, licensed-spectrum reliability for utility-scale metering, as in Module 2's worked example) → Utility Cloud Platform (time-series storage, Module 3, Section 3.3) → Demand-Forecasting Model → Grid Operations Dashboard.

Smart metering is the direct real-world instance of Module 2's own NB-IoT worked example: static devices, an eight-year-plus battery budget, and a need for licensed-spectrum reliability at national scale all point to NB-IoT as the natural communication choice, exactly as reasoned there.

7 Smart Transportation

Smart transportation connects vehicles, infrastructure, and fleet-management systems, and is unusual among the domains in this module in that many of its devices are **mobile** rather than fixed – a constraint that shapes both its communication and processing choices.

- **Vehicle tracking:** GPS modules (Module 1, Section 4.3) report vehicle location continuously, the foundation for fleet visibility and route optimisation.
- **Fleet management:** Combines location, fuel/battery status, and driver behaviour data to optimise routing, scheduling, and vehicle utilisation across a fleet.
- **Intelligent traffic systems:** Vehicles and infrastructure share data to optimise traffic flow city-wide, overlapping directly with Section 5's smart traffic management.
- **Connected vehicles:** Vehicle-to-vehicle and vehicle-to-infrastructure communication (often over 5G, Module 2, Section 3.3) supports cooperative safety features such as collision warnings.
- **Accident detection:** Onboard sensors (accelerometers, impact sensors) automatically detect collisions and can trigger an emergency alert without driver intervention – directly analogous to the fall-detection systems of Section 3.
- **Predictive maintenance:** Vehicle sensor data feeds the same predictive- maintenance techniques introduced in Section 4, applied here to engines, brakes, and batteries rather than factory machinery.

Representative Pipeline

Onboard GPS/Impact Sensors → LTE-M / 5G (Module 2: mobility support, low latency for safety features) → Edge AI in Vehicle (Module 3, Section 4.4: instant collision response) → Fleet Cloud Platform → Dispatcher Dashboard / Emergency Services Alert.

Because vehicles move between cell towers, **LTE-M** – which, unlike NB-IoT, explicitly supports handover between towers (Module 2, Section 3.2) – or full 5G is preferred over LoRaWAN or NB-IoT here, and because collision detection shares the same millisecond-scale safety requirement discussed for both industrial shutdown (Section 4) and ECG alerting (Section 3), it too is handled at the edge, inside the vehicle itself, rather than depending on a cloud round trip.

8 Smart Home

Smart home applications bring IoT into individual residences, typically at much smaller scale than the other domains in this module (tens rather than thousands of devices), which changes the calculus on several of the trade-offs discussed above.

- **Smart lighting:** Lights are controlled automatically by occupancy, time-of-day, or voice command, and can be scheduled or dimmed for energy savings.
- **Smart security:** Door/window sensors and cameras monitor for unauthorised entry, often integrated with the intrusion-detection systems below.

- **Smart appliances:** Connected refrigerators, washing machines, and other appliances report status and can be controlled or scheduled remotely.
- **Energy management:** Home energy monitoring, similar in principle to Section 6 but at household scale, tracks and helps reduce individual appliance energy use.
- **Voice-controlled devices:** Voice assistants provide a natural-language interface to the other smart-home devices in this list, translating spoken commands into device actions.
- **Intrusion detection:** Motion sensors and cameras, often paired with computer-vision-based person detection (Module 3, Section 5.4), distinguish genuine intrusions from false triggers (pets, passing headlights).

Representative Pipeline

Motion/Door Sensors → Zigbee Mesh or Wi-Fi (Module 2: short range sufficient within a single home) → Home Hub (local edge processing, Module 3, Section 4.4) → Cloud (remote access, voice-assistant integration) → Mobile App / Automated Actuator (lock, light).

At this small scale, **Zigbee**'s mesh topology (Module 2, Section 2.2) is often favoured precisely because it was designed for exactly this kind of building-wide, fixed-device deployment, and much processing can remain on a local home hub rather than the cloud – partly for the low latency a security system needs, and partly so that core functions (locking a door, turning on a light) continue to work even during an internet outage, echoing the offline-operation advantage of edge computing raised in Module 3, Table 2.

9 Cross-Domain Synthesis

Table 1 draws the seven domains above together, making explicit how each domain's constraints (device scale, mobility, and latency sensitivity) drive a different point in the design space developed across Modules 2–4.

Domain	Typical Link	Edge vs Cloud	Dominant Security Concern	Analytics Focus
Smart Agriculture	LoRaWAN	Mostly cloud	Device authenticity	Irrigation scheduling
Healthcare	BLE	Edge for alerts	Confidentiality	Anomaly/emergency detection
Industrial (IIoT)	Wired/Wi-Fi	Edge for safety	Availability	Predictive maintenance
Smart City	NB-IoT/Zigbee	Fog aggregation	Data integrity	Real-time optimisation
Smart Energy	NB-IoT/LTE-M	Mostly cloud	Data integrity	Demand prediction
Transportation	LTE-M/5G	Edge for safety	Authenticity	Predictive maintenance
Smart Home	Zigbee/Wi-Fi	Local hub	Confidentiality	Occupancy/intrusion detection

Table 1: Cross-domain synthesis: the dominant Module 2 communication technology, Module 3 edge/cloud balance, Module 4 security priority, and analytics focus for each application domain.

9.1 A Consistent Design Question

Across every row of Table 1, the same underlying question determines the technology choice: *how time-critical is the response, and how constrained is the device?* A domain with a hard, safety-relevant latency requirement (industrial shutdown, vehicle collision response, cardiac emergency alerting) pushes processing to the edge regardless of the extra engineering cost; a domain with no such requirement (irrigation scheduling, demand forecasting) is free to centralise processing in the cloud, where more historical data and computational power are available. This is precisely the trade-off framework developed in Module 3, Section 4.2, applied consistently across every domain in this module.

Worked Example

Applying the synthesis. A new application monitors bridge structural sensors (strain gauges, accelerometers) across a national infrastructure network, where an anomaly could indicate imminent structural failure, sensors are fixed in place for years, and network coverage at bridge sites is inconsistent. Using Table 1 and the reasoning in Section 9.1 as a guide, propose (a) a likely communication technology and (b) where anomaly detection should run, with justification for each.

Solution. (a) A long-range, low-power, licensed-spectrum technology such as NB-IoT is a reasonable choice, matching the smart-energy and smart-city rows of Table 1: sensors are fixed and long-lived, and licensed spectrum offers more reliable coverage than unlicensed LoRaWAN at potentially remote bridge sites. (b) Anomaly detection should run at the edge, on or near the

sensor itself, following the same reasoning as industrial shutdown and vehicle collision detection in Table 1: a structural-failure warning is safety-critical and cannot depend on the inconsistent network coverage described in the scenario, so a cloud round trip cannot be relied upon for the initial detection, even though the reading can still be forwarded to the cloud afterward for longer-term structural-health analytics.

9.2 The Road to IoT 4.0

The seven domains above are also converging along a common technological trajectory, often described in four stages: **IoT 1.0** (simple connected things, little processing), **IoT 2.0** (cloud-connected things, the architecture of Module 3, Section 3), **IoT 3.0** (AI-enabled IoT, the analytics of Module 3, Section 5 and Module 4, Section 5), and **IoT 4.0** (Edge AI, 5G, digital twins, and increasingly autonomous systems, combining nearly every technique developed across this course). Each domain in this module currently sits at a different point along that trajectory – healthcare and industrial IoT, with their hard latency requirements, are furthest toward IoT 4.0's edge-AI end, while some smart-agriculture and smart-energy deployments remain closer to IoT 2.0/3.0 – but all seven are moving in the same direction.

Course Synthesis: How Every Module Connects Here

Every application in this module is simply a specific instantiation of the ideas developed across the whole course. The Perception/Network/Processing/Application architecture of Module 1 is the skeleton every domain in this module fills in differently. Module 2's menu of short- and long-range links, and its application protocols, supplies the answer to "how does this domain's data actually move?" – LoRaWAN for agriculture's remote fields, BLE for healthcare's wearables, LTE-M/5G for transportation's mobile vehicles. Module 3's hardware platforms and edge-to-cloud spectrum supply the answer to "where is this domain's data processed, and by what?" – edge AI for every safety-critical domain in Table 1, cloud analytics for the rest. Module 4's security requirements and threat model supply the answer to "what must this domain protect, and against what?" – confidentiality above all in healthcare, availability above all in industrial control. Having completed this module, you have not just learned seven applications: you have seen the full course applied, end to end, seven times over.

10 Solved Problems

Worked Example

Problem 1. A smart-city air-quality network has 15,000 sensors spread across a metropolitan area, each reporting once every ten minutes. Using the reasoning from Section 5 and Module 3, Section 4.1, explain why a fog-computing tier is likely to be included in this deployment's architecture rather than sending every reading directly to a central cloud.

Solution. At 15,000 sensors reporting every ten minutes, the aggregate volume of readings reaching a single central cloud endpoint grows directly with device count, exactly the bandwidth and scaling concern raised in Module 3, Section 4.1; a fog tier aggregating and pre-filtering data at the district level, as described in Section 5's pipeline, reduces this load before it reaches the central cloud, mirroring the smart-building worked example in Module 3.

Worked Example

Problem 2. Compare the edge-versus-cloud balance for (a) smart-agriculture irrigation decisions and (b) industrial machine shutdown, and explain the difference in terms of the design question posed in Section 9.1.
Solution. Irrigation decisions tolerate delay of minutes to hours without serious consequence, so processing can run in the cloud where more historical and weather data is available, as shown in Section 2's pipeline. Machine shutdown must occur within milliseconds of detecting a fault to prevent damage or injury, so it must run at the edge, as shown in Section 4's pipeline. The difference follows directly from Section 9.1's question: irrigation has no hard latency requirement, while machine shutdown is safety-critical.

Worked Example

Problem 3. A wearable fall-detection device for elderly care and a fleet-tracking device for a delivery van both need to report location and an event (a fall, or a collision) reliably. Identify one similarity and one difference in their communication technology needs, referencing Module 2.
Solution. Similarity: both need to report a safety-relevant event reliably and with low latency, since a delayed fall or collision alert could have serious consequences for the person or property involved. Difference: the wearable is stationary relative to its paired hub (worn on one person, typically near a home BLE/Wi-Fi hub), so a short-range technology such as BLE suffices; the delivery van moves continuously across a wide area and between cell towers, so it needs a mobility-supporting long-range technology such as LTE-M or 5G rather than a short-range link.

Worked Example

Problem 4. A smart-home security system's manufacturer proposes processing all motion-sensor video entirely in the cloud, with no local (edge) processing at all. Using Section 8 and Module 3, Table 2, identify one practical disadvantage of this design.
Solution. A cloud-only design would stop functioning during an internet outage – the home's motion detection and intrusion alerts would go entirely offline exactly when they might be most needed – whereas a design with local edge processing on a home hub can continue detecting intrusions and, where applicable, triggering local alarms even without internet connectivity, as Module 3, Table 2 identifies "can operate offline" as a specific advantage of edge computing over a cloud-only design.

11 Summary

Seven application domains – smart agriculture, healthcare, industrial IoT, smart city, smart energy, smart transportation, and smart home – each instantiate the same underlying Sensor → Network → Processing → Application/Actuator pipeline developed across Modules 1–3, but each selects a different point in the design space according to its own constraints. Device scale and remoteness favour LoRaWAN or NB-IoT for agriculture and energy; battery-powered, body-worn devices favour BLE for healthcare; mobility favours LTE-M or 5G for transportation; and building-scale, fixed deployments favour Zigbee for smart homes and cities. Safety-critical, latency-sensitive decisions – industrial shutdown, collision response, cardiac emergency alerting – are consistently pushed to the edge, while longer-horizon analytics such as demand forecasting and predictive maintenance run comfortably in the cloud. Security priorities likewise shift by domain, from healthcare's overriding

emphasis on confidentiality to industrial IoT's emphasis on availability. Together, these seven domains show the entire course – Modules 1 through 4 – applied in practice, and point toward the broader convergence of edge AI, 5G, and digital twins often described as IoT 4.0.

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